

Critical Thinking 7: Final Draft Final Research Paper

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ABSTRACT

Upon exploring the Final Capstone Project, this research paper investigates the application of Healthcare Predictive Data Analytics to formulate early prevention strategies regarding Diabetes Management. In utilizing the 2015 Behavioral Risk Factor Surveillance System (BRFSS) dataset consisting of 253,680 patient records and 22 variables, the study examines the predictive viability of both clinical indicators and socioeconomic factors. A comparative quantitative framework was utilized through SAS OnDemand for Academics, exploring Logistic Regression baseline models and advanced Random Forest machine learning. Conclusively, the evidence proved to reject both Null Hypotheses of the two Research Questions formulated for the Capstone project and demonstrated that Diabetes related indicators as Body Mass Index and socioeconomic determinants as Income both share a statistically significant predictive relationship with a diabetes diagnosis. The Capstone Project also concluded from the statistical and visual evidence that Healthcare Machine Learning algorithms would benefit from integrating social determinants identified as multidimensional to accurately and effectively flag at-risk patients prior to diagnoses.

INTRODUCTION

Upon exploring the Final Capstone Project deliverables due in Week Seven 'Final Research Paper' and Week Eight 'Oral and Power Point Presentation', the First Draft of the Final Research paper, explores the Healthcare industry, focussing upon the following focal points from all Critical Thinking Assignments. The Dataset in question, tools as well as techniques utilized throughout the Data Analytics Project were introduced cumulatively along with a Data Visual of the Meta data introduced in Critical Thinking Two and are fully integrated into the First Draft of the Research Paper. Required deliverables of Critical Thinking Three, addressing each Null and Alternate Hypotheses formulated, Ethical Considerations, and Research Methods were explored thoroughly and can be found evident in the following paper. The last section fully refers to the Literature Review produced to meet deliverables of Week Four. The Literally sources of the Research Paper provide both scope and justification to the Data methodologies explored, ethical considerations, and Research problems the paper addresses in full.

EXPLORING THE RESEARCH TOPIC

The application of Healthcare Predictive Data Analytics to formulate early prevention and diagnoses strategies regarding Diabetes Management are also examined relevant to prevalent health concerns as Diabetes Diagnoses. The impact as well as relationship of different variables to one another are found imperative to extract research insight and examine causations of chronic health diseases and which provide context to exploring the research project. Insights relevant demographic related variables as lifestyle choices

and socioeconomic factors is one correlation that could further scientific research. In examining the deliverables required of Week One, a proposal was compiled specific to introducing the topic of Capstone Project, the Dataset in question exploring the business problem at hand, and discuss ways Organizations may benefit from targeted research areas that in conjunction with Data Analytics which are projected to make varying advances in real world application.

As a well documented and prevalent health concern to impact millions of individuals, 'Diabetes' can be described to be a serious chronic disease that greatly plagues individuals each year and which operates as a disease where individuals lose the ability to effectively regulate levels of glucose in the blood. Teboul, A. (2021). Due to Diabetes work rapidly progressing, it is essential for the Healthcare Sector and companies involved as Hospitals, Medical and Speciality Clinics related to Diabetes Management to be able to implement proactive Healthcare related strategies. Medical advancements have been extensive with the use of Predictive Analytics however there still exists a significant issue regarding under and delayed diagnoses long after the irreversible changes have begun to impact the human body. To distinguish further concerns in Diabetes Management, Healthcare providers often lack high accessibility to data driven prediction models capable of flagging demographic factors prior to blood testing for Chronic conditions as Diabetes.

DATASET JUSTIFICATION

Once foods are broken down into varying sugars during the digestion period, the sugars become released into the blood stream of the diabetic individual for energy. Teboul, A.

(2021). Diabetes is further typically described in encompassing definition to be either by the body not making enough insulin or an inability to utilize the insulin as effectively as needed for healthy functionality, the result of failure leading to further health complications as heart disease, vision loss, lower limb amputation and Kidney diseases, all which are strongly correlated with high levels of sugars that exist in the blood stream for those with Diabetes. Teboul, A. (2021). The scale as well as scope of the prevalent disease was found a critical research area to explore and the premise of the Capstone Project.

The Centers for Disease Control and Prevention indicated as of 2018, 34.2 Million Americans had Diabetes during that time period and '88' Million had pre-Diabeties. The CDC further estimated '1 in 5' diabetics and an estimated '8 in 10 pre-diabetics were unaware of the risks associated with diabetes. Teboul, A. (2021). In justification of exploring the research topic further, in context Type 2 Diabetes are described to be most prevalent which varies by age, education, income, location, race, and other social determinants of health. Teboul, A. (2021).

The global Diabetes epidemics are described by another source to have reached substantial portions within an estimated '537 million adults currently impacted and which is projected to significantly increase within the next decade. Deng, M., Yang, R., Zheng, X., Deng, Y., & Jiang, J. (2025). Digital Health Technologies relevant to Artificial Intelligence (AI) holds a significant potential to address existing gaps in the ability to address challenges relevant to under-reporting, diagnosing, and inaccurately predicting patient outcomes. Specific to Diabetes Management, AI applications are commonly applied to screening and diagnosis where automated screenings tools take a

non-invasive image classification recognition approach to improve the identification of those individuals that are at high risk to be diagnosed with Diabetes. Deng, M., Yang, R., Zheng, X., Deng, Y., & Jiang, J. (2025).

Artificial Intelligence is also largely applied to Diabetes Management to prevent and predict the presence and likelihood of a person experiencing Diabetes. Machine Learning based risk prediction models commonly utilize multi dimensional data mining to identify high populations so as to implement early interventions strategies. Deng, M., Yang, R., Zheng, X., Deng, Y., & Jiang, J. (2025).

RESEARCH PROBLEM

Critical Thinking One introduced prevalent health concerns as Diabetes Diagnose and the impact of the chronic disease as well as explored the relationship of different variables to one another . These were introduced as imperative for extracting research insight and examined the causations of chronic health diseases. Insights relevant demographic related variables as lifestyle choices and socioeconomic factors were also discussed further explored in correlation to further scientific research.

When involving Diabetes, understanding and mitigating the impact of Socioeconomic Determinants Of Health (DOH) are described by one literature source to be critical due to disease prevalence, economic costs, and disproportionate population burden. (Hill-Briggs et al., 2020)

Further described by author Hill Briggs, text published in the Year 2020, the

differences in diabetes risk and outcomes can result from multiple contributors, including biological, clinical, and nonclinical factors.

(Hill-Briggs et al., 2020)

Author Hill-Briggs elaborated within the literature , a substantial body of scientific literature has demonstrated adverse impact of a particular type of difference, health disparities in diabetes (Hill-Briggs et al., 2020). Further discussed, there exists vast amount of health disparities research in the U.S. has examined disparities by race and ethnicity. (Hill-Briggs et al., 2020)

Internationally, the new source introduced in the Final Research Project, regarding the term 'health equity', it has traditionally been used to encompass the range of population inequalities resulting from demographic and economic characteristics, and this term is used increasingly in the U.S. (Hill-Briggs et al., 2020)

To provide further merit, the source was found to address 'healthy equity' necessitates an understanding of social and environmental factors that are estimated to be made up of '50%' to '60%' of health outcomes. These social and environmental factors collectively are known as SDOH.

(Hill-Briggs et al., 2020)

The Research Proposal was compiled specific to introducing the topic of Capstone Project, the Dataset in question explores the research problem at hand, and discusses ways Healthcare organizations may benefit from targeted research areas that in conjunction with Data Analytics are projected to make varying advances in real world application.

OVERVIEW OF STUDY

In reiterative focus, the industry selected during the Research Proposal phase of the project regards the Healthcare Sector emphasizes upon the benefits of applying Healthcare Predictive Data Analytics through strategic means, formulating early prevention and diagnoses strategies regarding Diabetes Management and relevant to prevalent health concerns as Diabetes Diagnoses. The impact of examining relationships of different variables to one another provide rich valuable insights and further correlating relationships between variables are concluded to be imperative in order to extract research insight in examining the causations of chronic related diseases specific to Diabetes emphasizing the context to the research problem at large.

As a well documented and prevalent health concern to impact millions of individuals, 'Diabetes' can be described to be a serious chronic disease that greatly plagues individuals each year and which operates as a disease where individuals lose the ability to effectively regulate levels of glucose in the blood. Teboul, A. (2021). Due to Diabetes work rapidly progressing, it is essential for the Healthcare Sector and companies involved as Hospitals, Medical and Speciality Clinics related to Diabetes Management to be able to implement proactive Healthcare related strategies. Teboul, A. (2021). In

order to further explore the Research Problem at hand specific to early prevention, diagnosis of diabetes, and improvement of Diabetes Management care, it was essential to examine what might be strong risk indicators and factors involved to accurately predict the likelihood of a patient being diabetic.

The Dataset in question 'Diabetes Health Indicators' originated from the Behavioral Risk Factor Surveillance System (BRFSS) conducted by the Centers for Disease Control and Prevention question and was examined specific to the Year 2015. The Behavioral Risk Factor Surveillance as an expansive health related telephone survey has operated over the last couple of decades to collect survey responses from over '400,000' Americans on a yearly basis regarding health related risks behaviors, chronic health conditions, and use of preventive services (inclusive of health screenings and doctor appointments). As an expansive health related telephone survey which examines demographic, health related risk behaviors, and chronic health conditions among adults in the United States, the dataset is identified as clean and aggregated. Consisting of '253,680' individual patient records and 22 distinct variables, the initial data handled required categorizing the dataset's binary mathematical variable types.

To provide further context, the Behavioral Risk Factor Surveillance System (also called BRFSS) is a survey that was developed and has been implemented since the Year 1984 and in partnership with the Center for Disease Control (CDC) and State Health Departments inclusive of the BRFSS Survey. State Health Access Data Assistance Center. (2024, August 20). The specified State Health Department participants involved refer to all fifty individual U.S States and United State territories. The Survey the Dataset originated from occurs annually and is described to be telephone based geared

toward U.S adults 18 years or older. State Health Access Data Assistance Center. (2024, August 20). Respondents of the survey are described to be contacted through methods as landlines and cellphones through random digit dialing and which questionnaire consists of three parts. State Health Access Data Assistance Center. (2024, August 20).

The Core Component of the Behavioral Risk Factor Surveillance System regards demographic related questions and inquiries regarding health related perceptions, conditions and behaviors. State Health Access Data Assistance Center. (2024, August 20). The core is additionally composed of a standard core of questions that all states ask survey participants each year and involve a rotating core of alternating questions. State Health Access Data Assistance Center. (2024, August 20). The second component consists of optional questions related to health conditions and additional risk factors which are optional and not mandatory for States to ask survey participants opposed to the first core component. State Health Access Data Assistance Center. (2024, August 20). The third and last component of the Behavioral Risk Factor Surveillance System allows States to add questions to their own questionnaires in order to gain specific insights about a topic or population. State Health Access Data Assistance Center. (2024, August 20).

Essential information collected by the BRFSS refers to 'Health Risk Behaviors', 'Chronic Health Conditions' and 'Use of Preventive Services'. Health Risk Behaviors address questions related to Tobacco use, alcohol consumption, exercise, and sleep. State Health Access Data Assistance Center. (2024, August 20). Chronic Health Condition related questions specify Cardiovascular diseases, Diabetes, Asthma, and Depression.

State Health Access Data Assistance Center. (2024, August 20). Preventive related Service questions specify questions related to Cancer screenings, vaccinations, and routine doctor as well as dentist checkups.

'Diabeties_012' is one identified variable selected as the target variable of the dataset 'Diabetes Health Indicators' for exploration and conducting further Data Analysis in further sections of the Final Capstone project. The Binary variables categorized involved converting Yes or No related response for high Blood Pressure or Smoking Status), Ordinal related variables (as income and age variables) education levels, and variables labeled as interval or continuous in data types. One focal point may emphasize calculated variables as 'Body Mass Index' and 'PhysicalHealthDays' identified as poor by Survey participants.

The Behavioral Risk Factor Surveillance System 2015 Dataset examined directly aligns with the Organizational research problem outlined in the initial research proposal formulated in Critical Thinking One. Healthcare Organizations such as Clinics, Diabetes treatment facilities, and Health insurance companies often lack necessary accessibility to predictive analytic tools that are capable of identifying at-risk patients early on or prior to treatment. In modern, current time, Diabetes diagnoses are a common chronic illness that requires invasive, time consuming and often financially costly treatment.

The BRFSS Diabetes Indicators Dataset is composed entirely of accessible data and by utilizing this specific type of readily available Datasets, organizations involved specifically to the Healthcare field can accurately address a vital need for accessible Predictive Data Analytic tools that are capable of identifying patients that are flagged as

high risk in examining both demographic and lifestyle related data. By correlating trends and variables among the dataset, such extracted insights are capable of allowing Healthcare Organizations as Clinicals, Hospitals and Health Insurance companies target company intervention strategies.

In order to meet the required deliverables of the Final Capstone project, Critical Thinking Two explores the Meta Data categorized into a 'Feature Map'. Originally a Data Dictionary was considered, however a Feature Map was found an adequate alternative to visually displaying the role of each variable, name of each variable, Data Type, and a brief description to provide further context to the Data explored. As Predictive Modeling relies greatly on existing correlationsing relationships between applicable factors, the 22 variables present were illustratively grouped into four primary predictive domains that consistent of the 'target variable', 'clinical and biometric indicators', 'life style behaviors' and 'socioeconomic determinants'.

Also to formulate the 'Categorization Table' , Google Sheets was utilized as an alternative to Microsoft Excel and documented in the screenshot integrated below. The Predictive Target variable is identified as 'Diabeties_012' and the Data Type identified as Nominal. In categorical conversation, '0' equates to 'No Diabetes', '1' converts to 'prediabetes', and '2' converts to 'Diabetes'. Clinical and Biometric predictive variables involved 'HighBP', 'Highchol', 'BMI', 'GenHlth', 'MentlHlth/PhysHlth', 'DiffWalk', and 'Stroke/Heart disease'. Specific lifestyle related behaviors, primary variables involved 'smoker', 'HvyPhysActivity', 'fruits/veggies', and 'HvyAlcoholConsump'. The last predictive domain examined socioeconomic factors of the Dataset refers to variables as 'Age', 'Education', 'Income', 'AnyHealthcare', and 'NodoctbcCost'.

Upon exploring the Final Capstone Project 'deliverables due both in Week Seven 'Final Research Paper' and Week Eight 'Oral and Power Point Presentation', Critical Thinking Two explored objectives relevant to the Dataset in question, comprehensively introducing the origins of the Dataset, Data Analytical tools to be implemented and technique to be exploration specific to Logistic Regression and Random Forest Algorithms. A Model Visual specific to creating a MetaData Categorization Table was also produced in a Google Spread Sheet to provide a full categorization of predictive domains and inclusive variables.

RESEARCH QUESTIONS AND HYPOTHESES

In response to the well documented and prevalent health concern plaguing millions of individuals each year, 'Diabetes' can be described to be a serious chronic health disease. Due to Diabetes work rapidly and progressively, it is essential for the Healthcare Sector and companies involved as Hospitals, Medical and Speciality Clinics related to Diabetes Management to be able to implement proactive Healthcare related strategies. Medical advancements have been extensive with the use of Predictive Analytics however there still exists a significant issue regarding under and delayed diagnoses long after the irreversible changes have begun to impact the human body. To distinguish further concerns in Diabetes Management, Healthcare providers often lack high accessibility to data driven prediction models capable of flagging demographic factors prior to blood testing for Chronic conditions as Diabetes.

Critical Thinking Three explored two formulated research questions justified by the prevalence and health concern the chronic disease Diabetes has in modern day society.

An overview of the hypothesis and introduction of both the Null and Alternate Hypotheses are also explored in the sections below as well as methodology to be utilized specific to Statistical tests relevant to each Hypotheses formed and Ethical considerations that arose. The scale as well as scope of the prevalent disease are a critical research focal point that act as premise of the Capstone Project.

As a well documented and prevalent health concern to impact millions of individuals, 'Diabetes' can be described to be a serious chronic disease that greatly plagues individuals each year and which operates as a disease where individuals lose the ability to effectively regulate levels of glucose in the blood. Teboul, A. (2021).

Due to Diabetes work rapidly progressing, it is essential for the Healthcare Sector and companies involved as Hospitals, Medical and Speciality Clinics related to Diabetes Management to be able to implement proactive Healthcare related strategies. Medical advancements have been extensive with the use of Predictive Analytics however there still exists a significant issue regarding under and delayed diagnoses long after the irreversible changes have begun to impact the human body.

To distinguish further concerns in Diabetes Management, Healthcare providers often lack high accessibility to data driven prediction models capable of flagging demographic factors prior to blood testing for Chronic conditions as Diabetes. Once foods are broken down into varying sugars during the digestion period, the sugars become released into the blood stream of the diabetic individual for energy. Teboul, A. (2021). Diabetes is further typically described in encompassing definition to be either by the body not making enough insulin or an inability to utilize the insulin as effectively as needed for

healthy functionality, the result of failure leading to further health complications as heart disease, vision loss, lower limb amputation and Kidney diseases, all which are strongly correlated with high levels of sugars that exist in the blood stream for those with Diabetes. Teboul, A. (2021).

The scale as well as scope of the prevalent disease was found a critical research area to explore and the premise of the Capstone Project. The Centers for Disease Control and Prevention indicated as of 2018, 34.2 Million Americans had Diabetes during that time period and '88' Million had pre-Diabetics. The CDC further estimated '1 in 5' diabetics and an estimated '8 in 10' pre-diabetics were unaware of the risks associated with diabetes. Teboul, A. (2021). In justification of exploring the research topic further, in context Type 2 Diabetes are described to be most prevalent which varies by age, education, income, location, race, and other social determinants of health. Teboul, A. (2021).

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With the primary objective of the Capstone project to examine and utilize Predictive Analytics, the Hypotheses are formulated with the objective in mind to examine the transition of Diabetes Management from a reactive approach to a proactive one which emphasizes diabetes health indicators. In the context of analysis, the '2015 Behavioral Risk Factor Surveillance System (BRFSS) Dataset, two research questions are explored along with each Null and Alternate Hypotheses in examining the chronic health issue of Diabetes.

Research Question One:

To what extent can lifestyle and non-invasive clinical indicators (such as BMI, Physical Activity, and Smoking status) accurately predict the likelihood of Diabetes utilizing Machine Learning?

- **Null Hypothesis:** The Null Hypothesis articulates that both Lifestyle related and non-invasive Clinical indicators do not have a statistically significant predictive relationship with leading indicators of a person becoming diagnosed with the chronic health concern, Diabetes.
- **Alternate Hypothesis:** The Alternate Hypothesis in contrast articulates that both

Lifestyle and None invasive Clinical share a statistically significant predictive relationship with leading indicators of a person becoming diagnosed with Diabetes.

Research Question Two:

How do socioeconomic related determinants such as income level, and education) influence the Predictive probability independent of an individual's physiological health (specific to damage that occurs to a person's cardiovascular system, nervous system and kidneys.)

- **Null Hypothesis:** The Null Hypothesis formulated regarding Research Question Two articulates socioeconomic related determinants as Income Level and Education do not have a statistically significant impact or hold influence to the predictive probability of a person becoming diagnosed with Diabetes independent of physiological health related indicators.
- **Alternate Hypothesis:** The Alternate Hypothesis states socioeconomic related determinants as income level and education do statistically influence the predictive probability of a Diabetes diagnosis independently of physiological health indicators and which demonstrate that non clinical related and social factors are statistically significant in its relevance to predictors of the chronic health concern of Diabetes.

LITERATURE REVIEW

The foundation of ensuring any robust Predictive Model lies in the structural integrity of

its underlying data. Public Health research is backed by the Centers for Disease Control and Prevention's 2015 Behavioral Risk Factor Surveillance System which represents a Gold Standard for Healthcare related modeling. In accordance with the State Health Access Data Assistance Center, the BRFSS Dataset is described to act as an essential tool for Healthcare researchers to utilize due to both its vast scale and comprehensive inclusive use of non-clinical variables. State Health Access Data Assistance Center. (2024, August 20). The Dataset's viability for Machine Learning(ML) is further justified by the vast adoptive use and study by the wider community of Data Scientists. The author Teboul extensively utilized adaptation of the Behavioral Risk Factor Surveillance System Health indicators dataset and described it to demonstrate specific survey data is structurally optimized for training and validating complex classification. Teboul's (2021)

The transition from reactive Clinical diagnosis to proactive disease prediction relies heavily on the evolution of statistical modeling. Foundational literature actively supports the deployment of advanced machine learning over traditional diagnostic frameworks. In a benchmark study, Author Xie published a study which demonstrated that machine learning algorithms regarding Decision Trees and Random Forests, provide a reliable framework for predicting Type 2 Diabetes. The study itself was based on survey participant responses rather than invasive Clinical lab tests. Xie et al. (2019). Building upon this, modern comparative analyses conducted consistently validate the advantages of superiority of ensemble methods over baseline type models. Authors Rajendra and Latifi published in the Year 2025 one paper which concluded that while traditional Logistic Regression provides vital interpretability and ensemble machine learning methods consistently yield superior overall accuracy, there exists sensitivity,

and precision when healthcare data that is of the multidimensional type. Rajendra and Latifi (2021). In exploring the SAS Institute, the Organization detailed Random Forest models to generate an ensemble of decision trees from bootstrapped data samples and inherently reducing the risk of overfitting and effectively managing the non-Linear relationships present in complex survey responses. SAS Institute (n.d.) Together, the literature in speaking statistically justifies this Capstone Project's methodology of utilizing Logistic Regression specific to baseline Socioeconomic interpretability, followed by a Random Forest model to maximize compatibility and predictive sensitivity.

Clinical Predictive Models historically have relied on physiological related factors and metrics, such as Body Mass Index (BMI). However, more current research over the last decade has confirmed there exists a necessary inclusion of social determinants of health. One author of a scholarly article confirmed there exists a highly effective predictive model that must take a data-driven approach that incorporates multidimensional related variables, including lifestyle choices (such as smoking status, physical activity) and socioeconomic status (income and education) in order to successfully identify at-risk groups. Dinh et al. (2019). Integrating these complex variables into Machine Learning introduces a significant number of challenges which should be aware of when considering Healthcare Predictive Analytics, with emphasis on class imbalance. Due to the fact that the vast majority of the general population does not have diabetes, another scholarly article articulated Predictive Models that naturally skew toward the majority class, creating a false sense of high accuracy all the while failing to detect true positive cases. The article noted that applying rigorous data preprocessing and balancing techniques before training ensemble models is absolutely

critical for the efficient and accurate diagnosis of diabetes mellitus. Olorunfemi et al. (2025)

In exploring the response to the well documented and prevalent health concern plaguing millions of individuals each year, 'Diabetes' can be described to be a serious chronic health disease. Due to the nature of Diabetes where it works rapidly and progressively to determine the human body to varying degrees, it is both the rational and justification to why the Healthcare Sector as a whole with emphasis to Hospitals, Medical and Diabetes Management clinicals highly benefit from taking interest in proactive healthcare related strategies. Critical Thinking Four encompassed the Literature Review portion of the written portion of the Final Capstone Project due in Week Seven, exploring the context of each source selected to provide merit to the subject of Healthcare and Predictive Analytics.

RESEARCH DESIGN

Methodology

In order to ensure comprehensive testing takes place to establish the Hypotheses compiled specific to Research Question One and Research Question Two, the Capstone Project will act within a quantitative as well as comparative framework. The objective in doing so is to determine if the two Predictive Model algorithms selected can accurately identify Diabetes risk more effectively than traditional based diagnostic related statistics traditionally utilized in the Healthcare field. One Scholarly Article examined provided merit to the basis of the research methods utilized in the Capstone Project, capitalizing upon that Public Health records commonly support the reliability of

Machine Learning algorithms and are described to be optimal techniques for predicting the likelihood of a Diabetes diagnosis originating from the Dataset Behavioral Risk Factor Surveillance System. Xie, Z., Nikolayeva, O., Luo, J., & Li, D. (2019).

Statistical Methods

In order to accurately extract valuable information pertaining to the Behavioral Risk Factor Surveillance Survey (BRFSS), a number of statistical techniques and Data Analysis tools were selected to explore the selected research problem. The primary Data Analytics Software tool selected to perform data exploration, statistical analysis functions and data manipulation is 'SAS onDemand for Academics', known well as SAS Studio and selected due to its popular use in the Healthcare sector and relevant to the industry in question. In use of the SAS OnDemand environment, three primary Data Analytic techniques were selected to apply and explore the research problem regarding first the application of Descriptive Statistics, second the application of Predictive Modeling specific to Logistic Regression and Random Forest in type and third, applying Class imbalance techniques in order to remedy the risk of skewed predictive results the models may be capable of producing. Prior to compiling the Logistic Predictive Model as well as Random Forest Regression, a baseline will be established with the application of Descriptive Statistics within the SAS environment.

Upon implementing SAS procedures, the baseline established may pertain to the distribution of Demographic related variables and explore the correlation between continuous interval variables (as 'BMI') and prevalence of the target variable selected 'Diabetes' may be accurately examined. The Data Analytical techniques selected to

examine further throughout the Final Capstone project will involve formulating Predictive based Classification models. In exploring Data Analytical techniques further, implementing a multinomial Logistic Regression model was found appropriate to accurately determine the statistical probability of a patient falling into prediabetes or a diabetes classification category.

Prior to deploying the two models, preprocessing and class imbalance are fundamental to address in completing the Final Capstone Project. The Dataset in question must undergo in-depth pre-processing within the SAS OnDemand environment. The BRFSS is observed to be a significantly imbalanced dataset where the vast majority of respondents fall into the 'no diabetes' majority class and 'pre-diabetics' class. If the Models are trained in these skewed circumstances, the algorithm itself will be biased and conclusively predict the majority class exclusively ('no diabetes') and fail to detect true positive cases. To accurately remedy the concern of class imbalance, the methodology selected specific to the Final Capstone project is to address data balancing solutions. Another source explored potential solutions to class imbalance relaying diagnostic classifications prior to training and described doing so likely will improve the probability of detecting minority classes. Olorunfemi, B. O., Ogunde, A. O., Almogren, A., Adeniyi, A. E., Ajagbe, S. A., Bharany, S., Altameem, A., Rehman, A. U., Mehmood, A., & Hamam, H. (2025).

As a final evaluation of both Logistic Regression and Random Forest Algorithm applied within the SAS studio environment, evaluating the predictive performance of both models ensures valid comparisons and overall findings. Model evaluation is necessary and intended to be examined in reference to predictive sensitivity and specificity. In

exploring further Evaluation specific tools, Confusion Matrices and overall Area Under the Curve (AUC) can determine in completion of the Project if the target variable 'Diabetes' is able to be accurately predicted on an individual basis before minority classification categories become overlooked.

Dataset Limitations

While the Behavioral Risk Factor Surveillance Survey is a highly reliable dataset, its utilization of secondary survey data introduces inherent limitations to the data. The primary limitation is self-reporting bias. Because the data relies on telephone surveys rather than direct clinical lab results, respondents may misreport their weight, income, or physical activity. Secondly, the cross-sectional nature of the 2015 Behavioral Risk Factor Surveillance Survey captures a single snapshot in time opposed to the ability to track longitudinal related progression of a patient prior to diagnosis of Type 2 diabetes to the diagnosis itself. Relevant to the time frame this may occur over a several year period. Lastly, as a fundamental point to limitations, the BRFSS dataset lacks Data relevant to complex genetic or hereditary indicators. This is restrictive and is limiting the predictive algorithm in order to capture both behavioral and socioeconomic factors relevantly found to the research survey.

Ethical Considerations

While the benefits of predictive analytics have become clear in the last decade, the Literature Review conducted around the following scholarly article, strongly cautioned against the ethical risks of deploying artificial intelligence in Healthcare settings. Emphasized by another source of review, the author explored algorithmic bias as a

paramount concern to producing accurate statistical results. If models are trained on datasets which fail to adequately represent minority or low-income populations in performance, the resulting algorithms will generate skewed predictions that actively exacerbate systemic healthcare disparities. Hanna et al. (2025)

There is a critical need for early detection however the current medical diagnostic procedures are riddled with errors. The literature source identified a number of primary diagnostic related failures as false negatives (where a diabetic patient is incorrectly identified as healthy), false positives (where a healthy individual is misdiagnosed), and unclassified cases resulting from a lack of historical data extraction Hossain, A. K. M. F., et al. (2025). False-negative errors are further discussed and described as particularly detrimental in public health, due to the fact that they delay necessary clinical intervention. Also traditional Machine Learning methodologies have historically yielded only small margins of accuracy when investigating such diagnostic gaps. This provides justification towards altering from traditional statistical approaches toward more advanced statistical frameworks. Hossain, A. K. M. F., et al. (2025). Furthermore, ethical Predictive Modeling should prioritize the end based-user. The author Markham argued in further research that integrating predictive tools into clinical practice requires extreme transparency to maintain patient trust Markham (2025). Additionally in the case a Predictive Model operates as an unexplainable, it loses ethical justification and trust of patients. The scholarly article emphasized a demand that predictive models must not only be statistically accurate but also be interpretable, morally balanced, and well communicated into practice with full transparency in mind. Markham (2025).

The escalating prevalence of the chronic health disease, Diabetes or known formally as

'Diabetes Mellitus', requires a resilient and urgent shift toward Healthcare practices which promote a proactive, data-driven diagnostic approach. The proposed methodologies introduced establish an appropriate framework that operates rigorously to test the predictive efficacy between that of ensemble machine learning against traditional statistics and Data Analysis. By utilizing the Behavioral Risk Factor Surveillance Survey, the dataset in question, the application of in depth preprocessing is possible in order to handle class imbalance. With the theoretical and methodological foundations secured, the phase of the research addressed the execution of Data Analytics in SAS Studio. Doing so, evidence has been in order to address the individual hypothesis and formulated recommendations to apply Predictive Analytics with the purpose of improving the quality and proactiveness of patient care and diagnoses.

Explored and introduced to the second Draft of the Research Paper and provided to the section of 'Literature Review', one source emphasized upon ethical considerations in Predictive Analytics specific to proactive Healthcare treatment. When integrated into electronic health records (EHRs), Machine Learning tools are capable of improving healthcare processes as clinical workflows and reduce errors in diagnoses. Bemporad, J., De Domenico, F., & Pozzilli, P. (2026).

Bemporard and Pozzilli further described the clinical value of these tools is contingent upon their reliability, transparency, and requires the appropriate integration of ML tools into human decision-making. This emphasizes upon the need for a balanced interaction between clinician expertise and algorithmic support. Bemporad, J., De Domenico, F., & Pozzilli, P. (2026). In addition, despite technological advances, several ethical tensions persist. AI systems often operate as 'black boxes', lacking transparency in their

decision-making pathways. Bemporad, J., De Domenico, F., & Pozzilli, P. (2026). They further may perpetuate biases embedded in training data, leading to unequal treatment outcomes, especially among underrepresented populations. Bemporad, J., De Domenico, F., & Pozzilli, P. (2026).

Moreover, the potential for automation bias, where clinicians defer too readily to algorithmic outputs, raises concerns about erosion of clinical judgement . Bemporad, J., De Domenico, F., & Pozzilli, P. (2026).

Another Literature to explore the ethical considerations in Machine Learning, capitalized upon privacy in the age of big data and also emphasized upon both advantages as well as disadvantages pertaining to Artificial Intelligence and Machine Learning in Healthcare.

AI Algorithms are described in this context to ultimately make recommendations for treatment and also discuss the advantages of ability to make resource allocation decisions which require use of very large sets of data from healthcare settings. Price, W. N., II, & Cohen, I. G. (2019).

Without this data, algorithms are unable to be trained or in instances once the Machine Learning Model is trained—they are assessed on how they perform. Price, W. N., II, & Cohen, I. G. (2019). This said, ethical challenges are elaborated to apply to considerations as collecting the Healthcare data in question. Authors by Price and Cohen noted, without access to massive data repositories, machine learning algorithms are entirely unable to be trained, nor can they be accurately assessed on their performance metrics once deployed. (Price & Cohen, 2019).

Price and Cohen further explored another Ethical consideration not presented in the prior portion of the Research Project and which introduces issues that exist in Data collection and how it may not occur equitably. (Price & Cohen, 2019). As an ethical matter, data collection is described by the two authors to be ‘best justified as a kind of “bargain” struck between data sources and data users, recognizing this may encroach in some ways on your privacy’. (Price & Cohen, 2019). The consequence of equitably shown by predictive analytics can appear in data mining. Health data are described to have many of the same problems where marginalized populations are missing from not health data related reasons. (Price & Cohen, 2019).

STATISTICAL FINDINGS

In reiteration of exploring Research Question One: Analysis of Research Question 1 specific regarding Clinical and Lifestyle related indicators: To what extent can lifestyle and non-invasive clinical indicators (such as BMI, Physical Activity, and Smoking status) accurately predict the likelihood of Diabetes?

To provide a comprehensive, in-depth visual analysis, a Box Plot was first produced. The variable ‘Body Mass Index (BMI)’ was one which was emphasized upon and when identifying the median specific to ‘0’ equating to the Healthy category appears visibly lower than the median of ‘1’ representing the Prediabetes classification and ‘2’ representing a Diabetic diagnosis, the Box Plot produced that the higher the BMI, the

greater the increase the concentrated focus of those patients diagnosed with Diabetes from the Behavioral Risk Factor Surveillance Survey.

When examining statistical findings, the Random Forest visual produced, one of the primary variables focussed upon in the Dataset 'Body Mass Index'(BMI), was concluded to be located at the very top of the Barchart produced. The 'Gini Reduction Value' was provided evidence the variable 'BMI' was most relied upon by the Machine Learning model and to the most of all variables present in the Dataset to most accurately predict the chronic health concern Diabetes

The Null Hypothesis (H1) relevant to Research Question One stating both Lifestyle related and non-invasive Clinical indicators do not have a statistically significant predictive relationship with leading indicators of a person becoming diagnosed with the chronic health concern, Diabetes is rejected and the Alternate Hypothesis (Ha1) accepted. The p' -value' was emphasized as sound statistical evidence that non-invasive clinical indicators (specifically BMI and High Blood Pressure) share a statistically significant predictive relationship with a diabetes diagnosis.

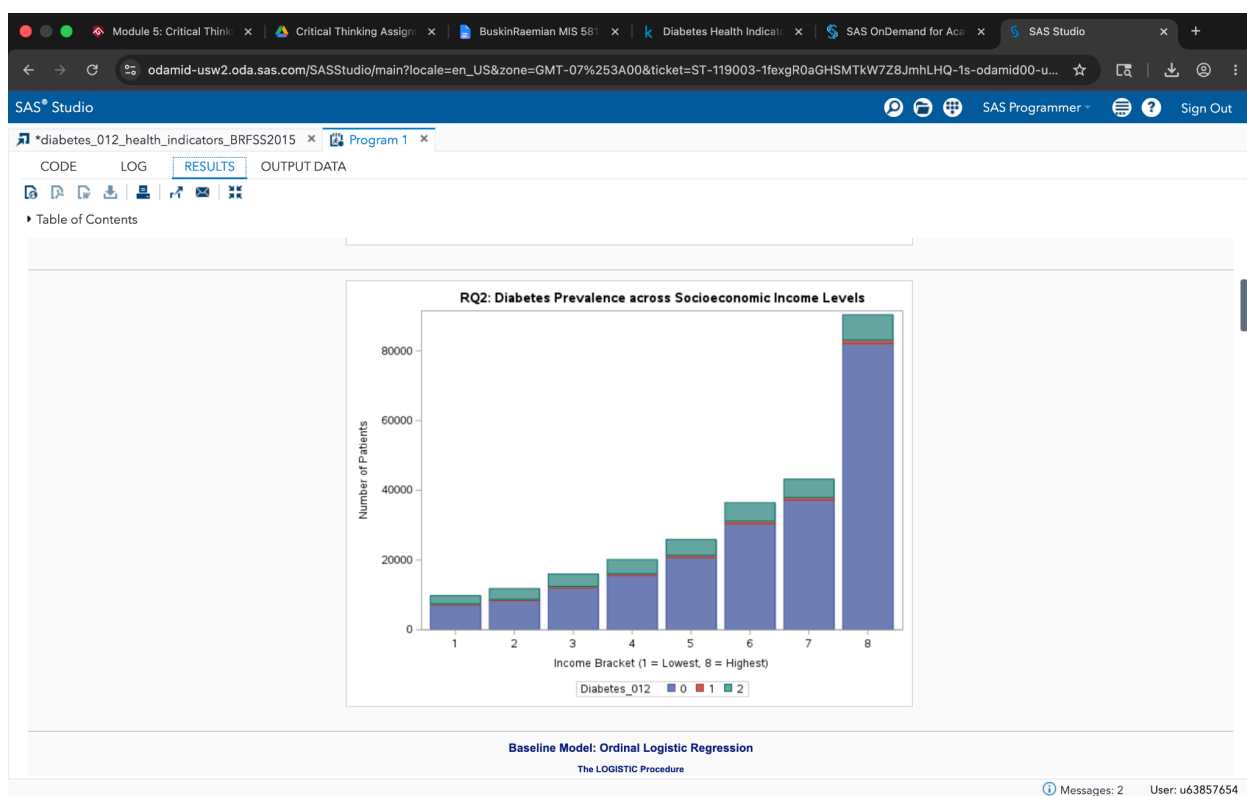
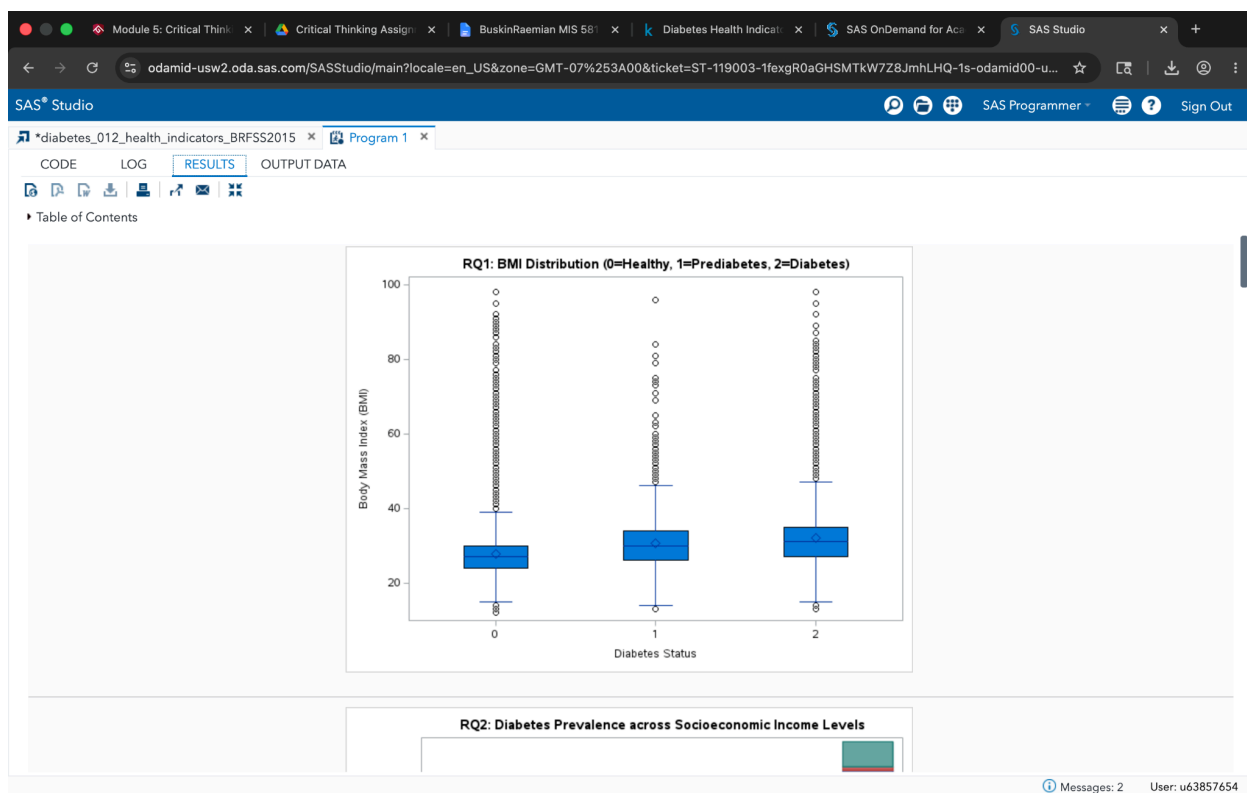
Upon examining the statistical and visual findings of Research Question Two specific to exploring Socioeconomic related Determinants, the produced results confirmed a number of findings specific to the Prediabetic and Diabetic diagnoses.

The Barchart produced is located within the lower income related brackets of the visual produced Lower income brackets (represented as Level 1 through Level 3), the colored segments of the chart are compiled to represent a vast portion of the barchart opposed to higher income brackets identified as 'Level 8'.

Statistically observed, the Forest Model produced explored the variables of the Behavioral Risk Factor Surveillance Survey as 'Income' and 'Age' second highest in ranking to the variable BMI when considering strongest predictive features.

The Logistic Regression was concentrated to return a P-value of ' < 0.0001 ' relevant to the variables of both 'Income' and 'Education' levels. This appeared to indicate there was a statistical significance independent of physiological related factors. The Null Hypothesis (H_0) is rejected in these circumstances and the Alternate Hypothesis (H_a) accepted. In reiteration, the alternate hypothesis specific to the second Research Question articulated socioeconomic related determinants as income level and education do statistically influence the predictive probability of a Diabetes diagnosis independently of physiological health indicators, demonstrating that non clinical related and social factors are statistically significant in its relevance to predictors of the chronic health concern of Diabetes. Further the Predictive Models formulated confirmed that socioeconomic related factors have a statistically influenced the probability of a diabetes diagnoses.

MODEL VISUALS AND DOCUMENTATION



SAS Studio

CODE LOG RESULTS OUTPUT DATA

Table of Contents

Baseline Model: Ordinal Logistic Regression

The LOGISTIC Procedure

Model Information	
Data Set	WORK.IMPORT
Response Variable	Diabetes_012
Number of Response Levels	3
Model	cumulative logit
Optimization Technique	Fisher's scoring

Number of Observations Read	253680
Number of Observations Used	253680

Response Profile		
Ordered Value	Diabetes_012	Total Frequency
1	0	213703
2	1	4631
3	2	35346

Probabilities modeled are cumulated over the lower Ordered Values.

Class Level Information						
Class	Value	Design Variables				
Smoker	0	0				
	1	1				
PhysActivity	0	0				
	1	1				
Education	1	0	0	0	0	0
	2	1	0	0	0	0
	3	0	1	0	0	0
	4	0	0	1	0	0
	5	0	0	0	1	0
	6	0	0	0	0	1

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SAS Studio

CODE LOG RESULTS OUTPUT DATA

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Probabilities modeled are cumulated over the lower Ordered Values.

Class Level Information						
Class	Value	Design Variables				
Smoker	0	0				
	1	1				
PhysActivity	0	0				
	1	1				
Education	1	0	0	0	0	0
	2	1	0	0	0	0
	3	0	1	0	0	0
	4	0	0	1	0	0
	5	0	0	0	1	0
	6	0	0	0	0	1
Income	1	0	0	0	0	0
	2	1	0	0	0	0
	3	0	1	0	0	0
	4	0	0	1	0	0
	5	0	0	0	1	0
	6	0	0	0	0	1
	7	0	0	0	0	0
	8	0	0	0	0	0

Model Convergence Status	
Convergence criterion (GCONV=1E-8) satisfied.	

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
196.5603	19	<.0001

Model Fit Statistics	
-2 Log Likelihood	

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Criterion	Intercept Only	Intercept and Covariates
AIC	249702.94	205337.16
SC	249723.83	205556.48
-2 Log L	249698.94	205295.16

Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	44403.7824	19	<.0001
Score	41944.3087	19	<.0001
Wald	32450.9748	19	<.0001

Effect	DF	Chi-Square	Wald	Pr > ChiSq
BMI	1	5442.0363	0.1943	<.0001
HighBP	1	3040.2109	0.1943	<.0001
HighChol	1	2356.0252	0.1943	<.0001
Smoker	1	0.0018	0.9658	0.0071
PhysActivity	1	7.2485	0.0071	<.0001
Age	1	2972.4289	<.0001	<.0001
Education	5	71.2682	<.0001	<.0001
Income	7	238.6735	<.0001	<.0001
GenHlth	1	6532.3890	<.0001	<.0001

Parameter	DF	Estimate	Standard Error	Wald	Chi-Square	Pr > ChiSq
Intercept	0	1	6.6033	0.1943	1155.2561	<.0001
Intercept	1	1	6.7791	0.1943	1217.1632	<.0001
BMI	1	1	-0.0628	0.000852	5442.0363	<.0001

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CODE LOG RESULTS OUTPUT DATA

Table of Contents

Baseline Model: Ordinal Logistic Regression

The LOGISTIC Procedure

Model Information	Value
Data Set	WORK.IMPORT
Response Variable	Diabetes_012
Number of Response Levels	3
Model	cumulative logit
Optimization Technique	Fisher's scoring

Number of Observations Read		253680
Number of Observations Used		253680

Ordered Value	Diabetes_012	Total Frequency
1	0	213703
2	1	4831
3	2	35346

Probabilities modeled are cumulated over the lower Ordered Values.

Class	Value	Design Variables
Smoker	0	0
	1	1
PhysActivity	0	0
	1	1
Education	1	0 0 0 0 0
	2	1 0 0 0 0
	3	0 1 0 0 0
	4	0 0 1 0 0

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SAS Studio

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Baseline Model: Ordinal Logistic Regression

The HPFOREST Procedure

Performance Information	
Execution Mode	Single-Machine
Number of Threads	2

Data Access Information			
Data	Engine	Role	Path
WORK.IMPORT	V9	Input	On Client

Model Information		
Parameter	Value	
Variables to Try	3	(Default)
Maximum Trees	50	
Actual Trees	50	
Inbag Fraction	0.6	(Default)
Prune Fraction	0	(Default)
Prune Threshold	0.1	(Default)
Leaf Fraction	0.00001	(Default)
Leaf Size Setting	1	(Default)
Leaf Size Used	2	
Category Bins	30	(Default)
Interval Bins	100	
Minimum Category Size	5	(Default)
Node Size	100000	(Default)
Maximum Depth	20	(Default)
Alpha	1	(Default)
Exhaustive	5000	(Default)
Rows of Sequence to Skip	5	(Default)

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SAS Studio

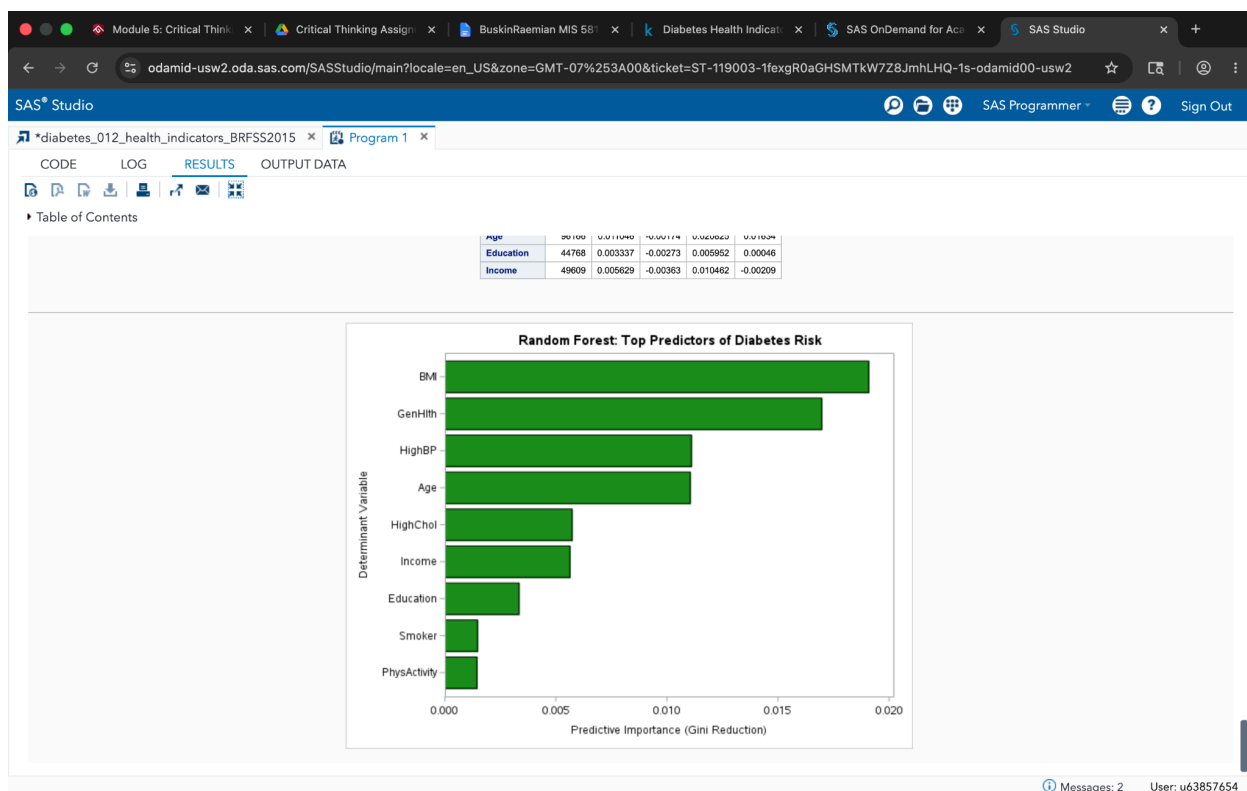
CODE LOG RESULTS OUTPUT DATA

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Baseline Fit Statistics	
Statistic	Value
Average Square Error	0.090
Misclassification Rate	0.158
Log Loss	0.492

Fit Statistics							
Number of Trees	Number of Leaves	Average Square Error (Train)	Average Square Error (OOB)	Misclassification Rate (Train)	Misclassification Rate (OOB)	Log Loss (Train)	Log Loss (OOB)
1	8909	0.0726	0.0858	0.146	0.163	0.655	1.149
2	18233	0.0677	0.0846	0.140	0.163	0.393	1.038
3	26526	0.0664	0.0832	0.138	0.162	0.353	0.920
4	33241	0.0660	0.0818	0.137	0.161	0.344	0.798
5	41604	0.0655	0.0808	0.137	0.160	0.339	0.721
6	48931	0.0654	0.0799	0.137	0.159	0.338	0.655
7	55616	0.0655	0.0792	0.137	0.158	0.338	0.605
8	64121	0.0653	0.0787	0.137	0.157	0.336	0.571
9	71553	0.0652	0.0782	0.137	0.157	0.336	0.539
10	80583	0.0650	0.0779	0.136	0.157	0.335	0.516
11	87946	0.0649	0.0776	0.136	0.156	0.335	0.494
12	95338	0.0649	0.0773	0.136	0.156	0.334	0.480
13	102965	0.0649	0.0771	0.136	0.156	0.334	0.468
14	111029	0.0648	0.0770	0.136	0.155	0.334	0.459
15	119438	0.0647	0.0768	0.136	0.155	0.333	0.452
16	127426	0.0647	0.0767	0.136	0.155	0.333	0.446
17	136309	0.0646	0.0766	0.136	0.155	0.333	0.441
18	144232	0.0646	0.0765	0.136	0.155	0.333	0.436
19	151485	0.0646	0.0763	0.136	0.154	0.333	0.432
20	159197	0.0646	0.0763	0.136	0.154	0.333	0.430

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SAS CODE EXECUTED

```
PROC SGPLOT DATA=WORK.IMPORT;
```

```
VBOX BMI / CATEGORY=Diabetes_012 FILLATTRS=(COLOR=CX0078D7);
```

```
TITLE "RQ1: BMI Distribution (0=Healthy, 1=Prediabetes, 2=Diabetes)";
```

```
YAXIS LABEL="Body Mass Index (BMI)";
```

```
XAXIS LABEL="Diabetes Status";
```

```
RUN;
```

```
/* Visual 2: Income Level Impact */
```

```
PROC SGPLOT DATA=WORK.IMPORT;
```

```
VBAR Income / GROUP=Diabetes_012 GROUPDISPLAY=STACK;
```

```
TITLE "RQ2: Diabetes Prevalence across Socioeconomic Income Levels";
```

```
XAXIS LABEL="Income Bracket (1 = Lowest, 8 = Highest)";
```

```
YAXIS LABEL="Number of Patients";
```

```
RUN;
```

```
PART 2: TRADITIONAL BASELINE MODEL (LOGISTIC REGRESSION
```

```
ODS GRAPHICS ON;
```

```
PROC LOGISTIC DATA=WORK.IMPORT;
```

```
CLASS Smoker (PARAM=REF REF='0')
```

```
PhysActivity (PARAM=REF REF='0')
```

```
Education (PARAM=REF REF='1')
```

```
Income (PARAM=REF REF='1');
```

```
MODEL Diabetes_012 = BMI HighBP HighChol Smoker PhysActivity Age Education  
Income
```

```
TITLE "Baseline Model: Ordinal Logistic Regression";
```

```
RUN;
```

ODS GRAPHICS OFF;

PART 3: ADVANCED ENSEMBLE MACHINE LEARNING (RANDOM FOREST)

PROC HPFOREST DATA=WORK.IMPORT MAXTREES=50 SEED=12345;

TARGET Diabetes_012 / LEVEL=NOMINAL;

INPUT BMI Age GenHlth / LEVEL=INTERVAL;

INPUT HighBP HighChol Smoker PhysActivity Education Income / LEVEL=NOMINAL;

ODS OUTPUT VariableImportance=WORK.VarImp;

RUN;

PROC SGPLOT DATA=WORK.VarImp;

HBAR Variable / RESPONSE=Gini CATEGORYORDER=RESPDESC

FILLATTRS=(COLOR=CX228B22);

TITLE "Random Forest: Top Predictors of Diabetes Risk";

XAXIS LABEL="Predictive Importance (Gini Reduction)";

YAXIS LABEL="Determinant Variable";

RUN;

RECOMMENDATIONS

Healthcare Organizations specific to Hospitals, specialty Diabetes treatment clinics, and

Health Insurance companies are recommended to provide frequent updates to their organizations of their predictive risk models in order to assess the potential weight socioeconomic factors . Machine Learning Algorithms are recommended to carefully evaluate variables as 'income' and 'education' coinciding traditional clinical metrics such as BMI (focussed upon as a primary driver of Diabetes) to accurately capture the full scope of a Patient's likelihood of risks.

As emphasized upon throughout the Literature Review, the application of such Machine Learning Models and Artificial Intelligence are recommended to be deployed with one hundred percent transparency. Healthcare organizations such as Hospitals, Speciality Diabetes Treatment Clinics, and Pharmaceutical Companies are also recommended to routinely perform audits and examine the specific Machine Learning Algorithms deployed so as to ensure that any predictive weight regarding specific variables as income does not risk the result of algorithmic bias or occurrence of false negatives.

CONCLUSION

Upon exploring the Final Capstone Project deliverables due in Week Seven 'Final Research Paper' and Week Eight 'Oral and Power Point Presentation', the First Draft of the Final Research paper, explored the Healthcare industry, focussing upon the following focal points from all Critical Thinking Assignments. The Dataset in question, tools as well as techniques utilized throughout the Data Analytics Project were introduced cumulatively along with a Data Visual of the Meta data introduced in Critical Thinking Two and were fully integrated into the First Draft of the Research Paper. Required deliverables of Critical Thinking Three, addressing each Null and Alternate

Hypotheses formulated, Ethical Considerations, and Research Methods were explored thoroughly and could be found evident throughout the First Draft of the Research Paper. The last section fully referred to the Literature Review produced to meet deliverables of Week Four. The Literally sources of the Research Paper provide both scope and justification to the Data methodologies explored, ethical considerations, and Research problems the paper addresses in full. One dedicated section of the second draft relays a series of documented visuals that were produced to provide both merit and context to the statistical analysis. A section of recommendations were elaborated upon for Healthcare Organizations as Hospitals, Diabetes treatment clinicals, and Pharmaceutical companies. The last and final edit made to the second draft of the Final Research Paper expanded upon the ethical considerations of Machine Learning and Artificial Intelligence in Healthcare Predictive Analytics. To provide final justification two new sources were provided to the last and Final Draft of the Data Analytics Research Paper exploring the chronic Health concern Diabetes.

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